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Krishanu Karun*

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Abstract

We present the Parametric Trotter Expression (PTE) family of operator primitives — Global PTE, Product PTE, and Local PTE (LPTE) — and show that they form a hierarchy of increasingly noise-resilient circuit ansätze grounded in the Continuous Topological Interpretation (CTI) of quantum circuits. CTI reframes quantum circuits as continuous landscapes embedded in an extended parameter space, where entanglement acts as manifold enlargement and noise as geometric deformation. Building on this foundation, we develop TOPAZ, a topology optimization framework that pre-tunes ansatz parameters before hardware execution. TOPAZ combines a tensor-dot noise engine ($O(2^N)$ per term, 3,838× speedup over dense Kraus maps), a dual-MMA optimiser that decouples intensity and timing subspaces, and an explicit two-stage sparsification protocol that actively removes entangling gates (34% depth reduction, $\Delta\text{DM} = -0.0065$). Validation spans an 8-qubit H_2/H_4 VQE SOTA benchmark and a direct *cross-Hamiltonian transfer test* (H_4 VQE $\rightarrow$ 8-qubit TFIM) that demonstrates TOPAZ optimises for the explicit symmetries of the *target* Hamiltonian. On live `ibm_fez` (Heron r2) hardware we measure a one-way state-preservation fidelity of 57.58%. The two-way Loschmidt-echo bound $F_{\text{echo}} \approx (F_{1\text{-way}})^2 = (0.5758)^2 \approx 33.15\%$ caps the round-trip echo, yet the raw hardware echo reaches 17.27% (the residual reconciled with CNOT-unrolling depth of $U^\dagger$ and spectator crosstalk), while standard ZNE collapses to 5.78% — a direct confirmation of the pre-execution paradigm. The result is a **11.14×** hardware gain at 18σ over the product-Trotter baseline. Across every backend and noise model, the optimiser converges to a hardware-transferable fidelity floor — the structural well predicted by CTI — confirmed by a $6 \times 6 \times 10$ noise-model transfer matrix with transfer efficiency 0.9943.

Keywords: Continuous Topological Interpretation, Parametric Trotter Expression, Quantum Topology Optimization, Noise-Resilient Quantum Circuits, Structural Well, Method of Moving Asymptotes.

1 Introduction

Variational Quantum Algorithms (VQAs) [6, 12] are the leading approach for near-term quantum computing, but they face a fundamental tension: shallow circuits limit expressivity, yet coherence times demand shallow circuits. Existing quantum error mitigation (QEM) techniques — Zero-Noise Extrapolation (ZNE) [7], Clifford Data Regression (CDR) [8], Virtual Distillation (VD) [9] — all operate on circuit outputs after execution: they post-process, fit error models, or extrapolate from amplified-noise runs [16]. None modify the circuit structure itself.

This paper introduces TOPAZ, a framework that acts *before* the circuit runs. Given a parametric Trotter ansatz, TOPAZ tunes the continuous parameters — intensity weights ρ and timing angles τ — so that the noisy output is intrinsically closer to the ideal one before any post-execution mitigation. The framework is grounded in the Continuous Topological Interpretation (CTI) [1], which reframes quantum circuits as continuous landscapes rather than discrete gate sequences.

The paper’s contributions are organized along three axes:

PTE family of operator primitives (Secs. 3–5). We derive three variants of the Parametric Trotter Expression, forming a hierarchy of increasing structure:

- **Global PTE** (Sec. 3): the original $2^N \times 2^N$ weighted-sum ansatz with polar projection — a general operator framework that closes at $N \geq 4$ due to $O(N^2/4)$ gates per polar step.

- **Product PTE** (Sec. 4): a first-order Trotter approximation that compiles to 8 two-qubit gates on 8 qubits — a proof-of-concept that TOPAZ works.
- **Local PTE (LPTE)** (Sec. 5): per-pair PTE with per-pair SVD projection, 21 two-qubit gates, and 5 effective parameters per entanglement pair — the canonical ansatz and the zenith of the PTE family.

Tensor dot engine and noise models (Sec. 6). An $O(2^N)$ engine that applies k -local Kraus operators via tensor contraction rather than dense 2^{3N} superoperator construction (3,838× speedup). We describe the six noise models against which TOPAZ is validated: the legacy triple-channel correlated model, IBM’s calibrated FakeFcz and FakeMontrealV2 snapshots, and live `ibm_fez` hardware.

TOPAZ framework (Sec. 7). The framework that brings CTI to experiment: a dual-MMA optimiser with ρ/τ subspace separation, an explicit two-stage sparsification protocol that actively removes entangling gates, and algorithmic compatibility criteria. Key results include a 11.14× gain on live `ibm_fez` hardware (18σ), a $6 \times 6 \times 10$ noise-model transfer matrix with transfer efficiency 0.9943, and direct empirical observation of the structural well across every backend tested.

State preservation vs. echo measurement (Sec. 8.9). We give the complete one-way/two-way picture: a one-way state-preservation fidelity of 57.58% on live `ibm_fez`, the theoret-

ical two-way echo ceiling $F_{\text{echo}} \approx (F_{1\text{-way}})^2 \approx 33.15\%$, the measured raw echo 17.27% (with the CNOT-unrolling and spectator-crosstalk decomposition), and the failure of standard ZNE (5.78%) — together establishing that pre-execution ansatz optimisation succeeds precisely where post-processing extrapolation breaks down.

Domain specificity of optimised topologies (Sec. 8.6). A cross-Hamiltonian transfer test (H_4 VQE $\rightarrow$ 8-qubit TFIM) proves TOPAZ does not produce a generic reusable circuit: because both layouts use virtually identical gate counts (~ 264), the 278.9 mHa transfer penalty is attributable purely to structural mismatch — direct evidence that the optimiser tailors entangling channels to the target Hamiltonian.

2 Continuous Topological Interpretation

CTI [1] provides the theoretical foundation for TOPAZ. We summarise only what is needed for the rest of the paper.

2.1 Ambient space and polar projection

Standard quantum circuits operate in the unitary group $U(d) = \{U \in \text{Mat}_d(\mathbb{C}) : U^\dagger U = I\}$, a compact Lie group of dimension d^2 where $d = 2^N$. CTI embeds $U(d)$ as a constrained submanifold inside the full space of linear operators $\text{Mat}_d(\mathbb{C}) \cong \mathbb{C}^{d^2}$. A continuous circuit operator $A(\lambda)$ maps $\mathbb{R}^M \rightarrow \text{Mat}_d(\mathbb{C})$ without the unitarity constraint, allowing gradients to flow around barriers that would block a strictly unitary optimizer.

To execute operators on hardware, we project back to $U(d)$ via the polar decomposition. Every invertible $A \in \text{GL}(d, \mathbb{C})$ decomposes uniquely as $A = QP$ with Q unitary and $P = (A^\dagger A)^{1/2}$ positive definite [20, Sec. 8.1]:

$$\mathcal{P}_U(A) = A(A^\dagger A)^{-1/2}. \quad (1)$$

Q is the unique closest unitary to A in the Frobenius norm. This projection — **unitary regularization** — pins singular values back to 1.0, smoothing sharp wrinkles in the loss landscape and stabilizing gradient signals. The decomposition is unique if and only if A is invertible; in LPTE the per-pair operator $B_{(i,i+1)}$ is lower-bounded by the clipping floor $\max(\rho, 10^{-6})$, so $\sigma_{\min}(B)$ remains strictly positive and the SVD polar factor $U = B(B^\dagger B)^{-1/2}$ is uniquely defined, guaranteeing that the projection is deterministic and differentiable on the active block.

2.2 Entanglement as manifold enlargement

Without entanglement, operations are restricted to localized rotations within disjoint qubit clusters. For an N -qubit register partitioned into k clusters S_a , the set of preparable product states forms the unentangled manifold $\mathcal{M}_{\text{unt}}$ with dimension $\dim(\mathcal{M}_{\text{unt}}) = \sum_{a=1}^k d_{\text{local},a}$. Introducing entangling parameters $\lambda_{\text{ent},a}$ that bridge cluster boundaries yields the entangled manifold $\mathcal{M}_{\text{ent}}$:

$$\dim(\mathcal{M}_{\text{ent}}) = \sum_{a=1}^k (d_{\text{local},a} + d_{\text{ent},a}) > \dim(\mathcal{M}_{\text{unt}}). \quad (2)$$

Entanglement physically inflates the accessible parameter manifold, providing higher-dimensional navigation that the optimizer exploits to bypass local minima.

2.3 Noise as geometric deformation

Environmental noise (T_1 , T_2 , readout error) acts as a continuous vector field pulling the state landscape away from the target manifold. Instead of correcting the output, TOPAZ searches for a **structural well** — a region of the parameter manifold where the noise field and control gradients approximately cancel. By pre-tuning (ρ, τ) to sit inside this well, noise resilience is built directly into the circuit’s structural parameters.

We note an important scope caveat. While global quantum control landscapes are trap-free under full controllability [30], TOPAZ operates on a constrained subspace — the sparse LPTE submanifold, a finite gate set, and a noise-deformed objective. Our smoothness claims are therefore scoped to the local regularity conditions of the structural well, not to the global trap-free theorem.

3 Global PTE

The Parametric Trotter Expression (PTE) is a parameterized operator ansatz where the free variables correspond to gate intensities and timings rather than rotation angles. Unlike conventional VQA parameterisations that maintain strict unitarity at every step, PTE allows the operator to temporarily leave the unitary group, explore a larger manifold, and project back via polar decomposition — a feature that expands expressivity per gate [3].

3.1 Derivation from Trotter–Suzuki

For a Hamiltonian $H = \sum_{j=1}^M H_j$ decomposed into Hermitian (typically two-local Pauli) terms, the Trotter–Suzuki approximation [19, 13] splits time evolution into n slices:

$$e^{-iHt} = \lim_{n \rightarrow \infty} \left[\prod_{p=1}^n \prod_j \exp(-iH_j \delta) \right], \quad \delta = \frac{t}{n}. \quad (3)$$

At finite n the error scales as $O(t^2/n)$ [17].

We extend the Trotter decomposition by introducing two classes of design variables for each term j and layer k : intensity weights $\rho_{j,k} \in [0, 1]$ with $\sum_j \rho_{j,k} = 1$, and timing scaling factors $\tau_{j,k} \in [0, \tau_{\max}]$. The layer- k PTE operator is

$$B_k(\rho, \tau) = \sum_{j=1}^M \rho_{j,k} e^{-iH_j \delta \tau_{j,k}}, \quad (4)$$

and the full PTE operator is

$$A(\rho, \tau) = \prod_{k=1}^n B_k(\rho, \tau). \quad (5)$$

Route (A) — applying weights directly to the Hamiltonian, $e^{-i \sum_j \rho_j H_j \delta}$ — maintains unitarity by construction but limits expressivity to weighted Hamiltonians. Route (B), the PTE, creates weighted sums of exponentials. The distinction is fundamental: the sum of unitaries in Eq. (4) is generically non-unitary, which is the source of PTE’s expressive power. Structurally, the Trotter–Suzuki formula is a *product* of exponentials, $\prod_j \exp(-iH_j \delta)$ (a Lie-product formula), whereas the Linear Combination of Unitaries (LCU) framework [27] implements a *sum* $\sum_j \alpha_j U_j$ using ancillae and amplitude

amplification. Per layer, PTE adopts the LCU sum structure (Eq. (4)) and composes layers via a Trotter-like product (Eq. (5)); it is thus an LCU instance without ancillae, with unitarity restored geometrically.

3.2 Non-unitarity and polar projection

Remark 1 (Non-unitarity of LCU sums). A weighted sum of unitary operators is generically non-unitary. This is immediate: consider $B = \sum_j \alpha_j U_j$:

$$B^\dagger B = \sum_{j,k} \alpha_j^* \alpha_k U_j^\dagger U_k. \quad (6)$$

For $B^\dagger B = I$ we require the cross terms to cancel — a condition satisfied only when all but one $\alpha_j = 0$ or under special symmetric constructions. The observation is elementary but important: non-unitarity is not a flaw; the mathematical freedom to leave the unitary manifold enables exploration of operator spaces unreachable by traditional variational circuits.

We restore physical realizability via layer-wise polar projection. For each layer k :

$$U_k = B_k (B_k^\dagger B_k)^{-1/2}, \quad U_{\text{safe}}(\rho, \tau) = \prod_{k=1}^n U_k. \quad (7)$$

We compute U_k via SVD with a Newton–Schulz refinement step and a final determinant correction $U_{\text{safe}} \rightarrow U_{\text{safe}} \cdot (\det U_{\text{safe}})^{-1/d}$ to remove the global phase. These projections are instances of the Linear Combination of Unitaries (LCU) framework [27]: the SVD polar projection replaces LCU’s ancilla and amplitude-amplification machinery with a deterministic geometric retraction. In the language of qubitization [28], LCU+qubitization embeds $\sum_j \alpha_j U_j$ in a larger unitary via PREPARE/SELECT oracles; TOPAZ instead projects via $B(B^\dagger B)^{-1/2}$, trading probabilistic amplification for a direct manifold retraction. PTE therefore inherits LCU expressivity while remaining ancilla-free and executable on NISQ hardware.

3.3 Scope and limitations

The Global PTE is the original formulation [3], requiring $O(N^2/4)$ two-qubit gates per polar step and thus intractable for $N \geq 4$. Its inversion test on 8 qubits registers 0.004, equal to the noise floor $1/256 \approx 0.0039$. The structural lower bound on quantum-state decomposition permanently closes this line for multi-qubit systems. The Product PTE and LPTE variants that follow resolve this by introducing locality.

3.4 Objective functions

The optimizer minimises the noisy fidelity loss:

$$L_{\text{op}}(\rho, \tau) = 1 - \frac{1}{d^2} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{safe}}(\rho, \tau))|^2, \quad (8)$$

or the state-fidelity loss for a target state $|\psi_{\text{target}}\rangle$:

$$L_{\text{st}}(\rho, \tau) = 1 - |\langle \psi_{\text{target}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2. \quad (9)$$

A regularisation term keeps the pre-projection operator close to unitarity:

$$L(\rho, \tau) = L_{\text{prim}}(\rho, \tau) + \lambda \sum_{k=1}^L \|B_k^\dagger B_k - I\|_F^2, \quad \lambda = 10^{-3}. \quad (10)$$

4 Product PTE

The Global PTE’s $2^N \times 2^N$ construction cannot scale. The Product PTE replaces the dense matrix product with a first-order Trotter product that compiles term-by-term to hardware-native gates. For an N -qubit system with M Hamiltonian terms:

$$U_{\text{circ}}(\rho, \tau) = \prod_{j=1}^M e^{-i \rho_j H_j \delta \tau_j}, \quad (11)$$

where each factor is a k -local unitary ($k = 2$ for nearest-neighbour Pauli terms). The polar projection is applied per term via SVD, but the product structure eliminates the need for layer-wise $2^N \times 2^N$ reconstruction.

On 8 qubits with $M = 21$ nearest-neighbour Pauli terms (seven $\{XX, YY, ZZ\}$ triples), the Product PTE compiles to 8 two-qubit gates and achieves DM fidelity 0.6329 under FakeFey noise — a 9.35× gain over random baseline (0.0677), with the canonical trajectory shown in Fig. 2. The circuit has 42 parameters and no pruning mechanism (self-pruning is absent in the product form because the Trotter factors are individually unitary).

The Product PTE serves as a proof-of-concept: it establishes that the TOPAZ optimisation pipeline works and that the structural well is reachable even with a simple ansatz. However, its lack of self-pruning and limited fidelity make it a stepping stone to LPTE, which dominates it on every metric (Sec. 5).

5 Local PTE (LPTE)

The Local PTE (LPTE) is the zenith of the PTE family. It combines the expressivity of the Global PTE with the locality of the Product PTE by applying a per-pair PTE sum with per-pair SVD projection — retaining the weighted-sum structure that enables self-pruning while keeping the circuit depth tractable.

5.1 Construction

For each nearest-neighbour pair (q_i, q_{i+1}) in an N -qubit linear chain, we define a local PTE operator:

$$B_{(i,i+1)}(\rho, \tau) = \sum_{P \in \{XX, YY, ZZ\}} r_{i,P} e^{-i P \delta \tau_{i,P}}, \quad (12)$$

with intensities $r_{i,P} \in [0, 1]$ and timings $\tau_{i,P} \in [0, \tau_{\text{max}}]$, normalised so $\sum_P r_{i,P} = 1$. Each $B_{(i,i+1)}$ is projected via SVD to its nearest unitary $U_{(i,i+1)}$. The full LPTE operator is the product of these per-pair unitaries:

$$U_{\text{LPTE}}(\rho, \tau) = \prod_{i=0}^{N-2} U_{(i,i+1)}. \quad (13)$$

On 8 qubits with 7 nearest-neighbour pairs and 3 Pauli terms per pair, LPTE uses 21 two-qubit gates and 42 parameters — the same parameter count as Product PTE but 2.6× the gate count. The key difference is structural: the per-pair SVD projection in LPTE allows the optimizer to discover which pairs contribute to noise resilience and assign them nonzero intensity, while Product PTE’s purely unitary Trotter factors lack this selection mechanism.

Table 1. Full ablation study on the 8-qubit FakeFz benchmark using the LPTE ansatz (21 two-qubit gates in full configuration). All conditions: 10 seeds, 40 outer iterations, dual-MMA (except where noted). Significance vs. full TOPAZ: Bonferroni-corrected $\alpha = 0.05/36 \approx 0.0014$.

Condition	DM Fid.	2Q Gates	Active	Cohen’s d
Exact preparation (oracle)	0.924 ± 0.003	21.0	21.0	26.43***
Full TOPAZ (baseline)	0.737 ± 0.066	12.6	9.2	—
COBYLA (LPTE)	0.729 ± 0.069	15.6	13.6	0.12 (ns)
Single MMA	0.623 ± 0.166	19.9	19.9	0.80 (ns)
Adam (LPTE)	0.514 ± 0.209	16.5	14.8	1.02 (ns)
No polar decomposition	0.334 ± 0.170	16.7	15.8	2.14***
HEA (3-layer)	0.329 ± 0.110	28.0	37.2	3.97***
Product-only PTE	0.279 ± 0.175	16.7	16.0	3.09***
Random PTE	0.056 ± 0.031	21.0	21.0	8.69***

*** $p < 0.001$ (Bonferroni-passing at $\alpha_{\text{Bonf}} = 0.0014$). Cohen’s d vs. full TOPAZ.

Table 2. Parameter efficiency: PTE-family vs. HEA on 8 qubits under FakeFz.

Ansatz	Parameters	2Q Gates	DM Fid.
PTE (product)	42	8	0.6329
LPTE	42	21	0.9085
HEA (3-layer)	~ 96	28	0.1517

5.2 Self-pruning

The central empirical discovery is that LPTE **self-prunes** under noise. During optimization, the MMA solver drives the ρ gradients of low-value pairs below the clipping floor $\max(\rho, 10^{-6})$, where the gradient vanishes and the parameters remain frozen. Across 10 seeds under FakeFz, typically 8–24 of 42 (ρ, τ) dimensions survive, corresponding to ~ 9 of 21 entangling pairs actively contributing. Table 1 shows the 9-condition ablation confirming that self-pruning requires the full TOPAZ stack (PTE ansatz + polar projection + dual-MMA).

The key findings: Full TOPAZ (DM 0.737 ± 0.066) significantly beats product-only PTE (0.279 ± 0.175 , $p < 0.001$, $d = 3.09$), HEA (0.329 ± 0.110 , $p < 0.001$, $d = 3.97$), and the no-polar-decomposition variant (0.334 ± 0.170 , $p < 0.001$, $d = 2.14$). The polar projection is critical — removing it collapses fidelity by over half.

5.3 Parameter efficiency

Table 2 compares PTE-family parameter efficiency against HEA on 8 qubits under FakeFz. LPTE achieves 0.9085 DM fidelity with 21 two-qubit gates and 42 parameters, while HEA uses ~ 96 parameters and 28 gates for 0.1517 — a 6 $\times$ gap in fidelity per parameter.

The inductive bias is the reason: PTE’s parameterisation mirrors the Hamiltonian structure (each (ρ_j, τ_j) ties directly to a specific Hamiltonian term), so the ansatz starts in the correct functional region. An HEA of comparable gate count must expend parameters learning which qubits should be entangled at all.

6 Tensor Dot Engine and Noise Models

6.1 Tensor-dot noise engine

Applying a sequence of k -local Kraus operators to a dense density-matrix tensor normally requires constructing the full

Table 3. Per-evaluation runtime at $N = 10$ for one layered triple-channel Kraus map.

Engine	Time (s)	Complexity per term
Qiskit dense Kraus map	180.400	$O(2^{3N})$
Tensor-dot (this work)	0.047	$O(2^N)$
Speedup	$\sim 3838\times$	—

$2^{2N} \times 2^{2N}$ superoperator — complexity $O(2^{3N})$. The tensor-dot engine reframes ρ as a tensor with $2N$ indices and performs local contractions:

$$\rho'_{i'_S, i_{S^c}; j'_S, j_{S^c}} = \sum_a K_{i'_S, i_S}^{(a)} \rho_{i_S, i_{S^c}; j_S, j_{S^c}} K_{j_S, j'_S}^{(a)\dagger}, \quad (14)$$

where S is the active qubit set and S^c are spectator qubits. This reduces complexity to $O(2^N)$ per term, yielding a 3,838 $\times$ wall-clock speedup over the dense-Kraus reference at $N = 10$ (Table 3).

The engine plays different roles on different experiment paths. On the legacy triple-channel path, it handles both the ideal-state computation and noisy density-matrix evolution. On the FakeFz simulator path, its role is narrowed to the noiseless target state; noisy evaluations are delegated to Qiskit’s AerSimulator. On live `ibm_fez` hardware, the engine has no role — the physical backend executes transpiled gates directly.

This decomposition governs TOPAZ’s scaling. The tensor-dot engine is a ρ (density-matrix) simulator, so it is the exclusive cost driver on the *simulation* path, where scaling to hundreds of qubits is intractable ($O(2^N)$ per Kraus term remains exponential in N). On hardware, however, the optimiser never needs the engine: the physical backend returns measurement statistics directly, and the objective is a sum of single-qubit expectation values — a purely measurement-based cost that involves no density-matrix or Kraus-superoperator construction. Consequently, the same (ρ, τ) optimisation that is density-matrix-bound in simulation becomes *measurement-bound* on a real device, and its cost no longer grows with 2^N but only with the number of measurement settings and shots. Scaling to 100–200 qubits is thus impractical if TOPAZ is run as a simulator, but unproblematic on real hardware, where no density matrix or Kraus superoperator is ever formed.

6.2 Noise models

TOPAZ is validated against six distinct noise profiles, spanning the gap from idealized toy models to live hardware:

Triple-channel correlated (legacy). Three sequential channels: depolarising at $p = 0.005$, amplitude damping at $\gamma = 0.008$, phase damping at $\lambda = 0.012$, each with single-qubit and multi-qubit components. Multi-qubit correlations decay exponentially with qubit distance, $p_{\text{corr}}(d) = \alpha p e^{-d/\xi}$ with $\xi = 2.0$. The Kraus operators are standard [18]. This model captures the qualitative shape of NISQ noise without being tied to any specific device.

FakeFz. IBM’s calibrated noise snapshot of the live `ibm_fez` processor (Heron r2, 156 qubits), carrying per-gate error rates, qubit-specific T_1 and T_2 times, readout errors, and

the device coupling map [10]. All headline experiments use FakeFez as the noise model during optimisation.

FakeMontrealV2. A 27-qubit device-calibrated noise model used for the earlier port verification. A 3-iteration smoke test moved from 7.4% to 36.0% DM fidelity, confirming the optimiser finds productive descent directions quickly under full device noise.

Live ibm_fez hardware. The optimised (ρ, τ) from FakeFez are submitted to the live 156-qubit Heron r2 [11] processor for the inversion test. No re-optimisation is performed on hardware; the parameters transfer directly.

The six-model transfer matrix (Sec. 8.8) demonstrates that the structural well is noise-model agnostic: parameters optimised under any model perform at 99.4% of their on-model fidelity when evaluated under a different model.

7 TOPAZ Framework

7.1 Dual-MMA optimiser

The MMA algorithm [4, 5] constructs a strictly convex separable approximation of the objective at each iteration and solves it analytically subject to asymptotic bounds. It is globally convergent for smooth objectives under standard asymptote and move-limit management [4]; the dual-MMA simply interleaves two such convergent instances, one per subspace, preserving convergence because the subspaces are updated alternately with shared objective evaluations. The dual-MMA optimiser has two meanings that travel together:

Meaning 1: subspace separation. The intensity vector ρ and timing vector τ are heterogeneous parameter classes whose finite-difference step sizes differ by roughly four orders of magnitude ($\epsilon_\rho = 10^{-4}$ vs. $\epsilon_\tau = \pi/4$). A single MMA instance with a shared move-limit budget systematically misallocates the parameter-update budget. Dual-MMA runs two independent MMA solvers — one for ρ , one for τ — with their own asymptote, move-limit, and momentum state. The momentum-smoothed gradient:

$$\nabla_{\text{smooth}} = \alpha \nabla_{\text{cur}} + (1 - \alpha) \nabla_{\text{prev}}, \quad \alpha = 0.4, \quad (15)$$

stabilises convergence. Move limits adapt online based on parameter change magnitude.

Meaning 2: stacked modules across parameter classes. When a target algorithm carries an additional class of parameters (e.g., VQE rotation angles), TOPAZ replaces the classical optimiser with a second MMA module wrapped around its own small PTE ansatz. The two modules alternate, each treating the other’s parameters as fixed. This paper exercises two modules (structural + algorithmic); a third class slots in the same way.

7.2 Topology optimization via sparsification

The LPTE ansatz defines an entangling graph $G_{\text{LPTE}}(\rho, \tau) = (V, E)$ where each pair p contributes a 4-qubit unitary with normalised intensity $\kappa(p) = \sum_P r_{p,P}$. Passive self-pruning

Table 4. Multi-seed topology optimization benchmark ($\tau_\kappa = 0.05$, 5 seeds). F = full (Stage 1), S = sparse (Stage 3).

Seed	DM fidelity		2Q gates		Depth		Pruned
	F	S	F	S	F	S	
0	0.8487	0.8291	20	18	79	52	1
1	0.7456	0.7324	17	15	63	40	1
2	0.8910	0.8910	21	21	85	85	0
3	0.5220	0.5301	15	9	58	24	3
4	0.6802	0.6726	20	17	77	39	1

discovers a sparse subspace within the continuous parameter domain. We formalize an **active** two-stage protocol that removes entangling gates from the circuit:

Stage 1 — Continuous relaxation. Solve $\min_{\rho, \tau} \mathcal{L}(\rho, \tau)$ on E_{max} with $\rho \geq 0$ and no artificial floor. The optimizer naturally drives low-value pairs toward zero.

Stage 2 — Hard pruning. Remove pairs with $\kappa(p) < \tau_\kappa$ (pre-registered threshold $\tau_\kappa = 0.05$):

$$E_{\text{pruned}} = \{(q_i, q_{i+1}) \mid \kappa(p) \geq \tau_\kappa\}. \quad (16)$$

Stage 3 — Sparse re-optimization. Solve $\min_{\rho', \tau'} \mathcal{L}(\rho', \tau' \mid E_{\text{pruned}})$ from fresh random restarts on the reduced domain.

Across 5 seeds (Table 4), the pruned circuit achieves 14% gate reduction ($18.6 \rightarrow 16.0$), 34% depth reduction ($72.4 \rightarrow 48.0$), and mean DM fidelity loss of only -0.0065 ($0.7375 \rightarrow 0.7310$). Seed 3 removes three of seven pairs, cutting gates by 40% and depth by 59% while actually improving DM fidelity by $+0.0081$. The 34% depth reduction compounds on hardware, where fewer sequential gates mean shorter coherence exposure.

7.3 Algorithmic compatibility

Whether continuous parameter optimisation works on a given algorithm depends on four topological properties of its fidelity landscape: connectedness, compactness, continuity, and convexity. Algorithms satisfying all four — VQE, VQA, QAOA, parametrised quantum simulation — are natural TOPAZ targets. Grover’s algorithm as a whole violates continuity and convexity; the Grover oracle layer is compatible via a topological lift ($3.09\times$ gain), establishing the principle that non-differentiable algorithms can be protected by applying TOPAZ to their structural primitives.

8 Results

8.1 8-qubit trajectories under triple-channel noise

Configuration: 8 qubits, 9 nearest-neighbour Pauli terms, 18 parameters, 40-iteration cap. Pre-optimisation noisy fidelity ranges 0.07–0.70; post-MMA fidelity rises to 0.30–0.88, an absolute improvement of 18–23 pp (two representative runs, $0.172 \rightarrow 0.371$ and $0.233 \rightarrow 0.838$, are shown in Fig. 1). The ~ 47 pp spread in final fidelity reflects a multi-modal landscape (the dual-MMA navigates several local optima at different floor depths).

[Figure 1. Two representative 8-qubit optimisation trajectories from independent random seeds under triple-channel correlated noise. Run 1: 0.172 $\rightarrow$ 0.371; Run 2: 0.233 $\rightarrow$ 0.838.]

Figure 1. Two representative 8-qubit trajectories under triple-channel noise.

[Figure 2. Canonical FakeFez optimisation trajectory: steep early descent in 5–10 iterations, then convergence to stable floor at $F \approx 0.6329$ (Product PTE).]

Figure 2. Canonical FakeFez optimisation trajectory.

8.2 H₂ VQE on FakeFez

At 2 qubits (STO-3G basis, UCCSD ansatz), TOPAZ reduces the noisy UCCSD baseline of $\approx 70.6 \pm 2.3$ mHa to ≈ 12.8 mHa (5.5 $\times$ improvement). The result is consistent with the structural-well picture: the noise-resilient basin appears at 2 qubits as it does at 8 and 12.

8.3 Blind TOPAZ: combined H₂/H₄ VQE SOTA benchmark

This is the **Blind TOPAZ** benchmark: TOPAZ is used purely as a variational ansatz, and *blind* means the optimiser is ignorant of all ground-truth references — it never sees the target observable value, the target state, or the ideal state, and it receives no information about the QEM post-processing it will later be stacked with; the only signal it receives is the raw noisy energy (FCI references -1.13730604 Ha for H₂ and -2.138890 Ha for H₄ are used *only* for the final scoring of ΔE_{FCI} , never during optimisation). The LPTE ansatz is tested as a variational circuit for energy-based VQE under FakeFez. Three ansätze are compared — LPTE (per-pair PTE, optimised with the dual-MMA pipeline), HEA depth 3, and HVA $d = 2$ — on both the 4-qubit H₂ path [136, 143, 142, 141] and the 8-qubit H₄ path [125, 124, 123, 136, 143, 142, 141, 140]. All runs use ZNE, CDR, and VD post-processing across independent random seeds. Table 6 collects both molecules; H₂ uses 10 seeds, H₄ uses 3.

On H₂, LPTE plus ZNE+CDR reaches $\Delta E_{\text{FCI}} = 8.96$ mHa

Table 6. Combined H₂/H₄ VQE SOTA benchmark under FakeFez.

Molecule	Ansatz	ΔE_{FCI} (mHa) noisy	ΔE_{FCI} (mHa) noiseless	ZNE+CDR (mHa)	Conv.
H ₂ (4q)	LPTE (TOPAZ)	71.96 $\pm$ 20.9	26.72	26.16 $\pm$ 18.9	100%
H ₂ (4q)	HEA (depth 3)	53.47 $\pm$ 3.4	22.89	0.0*	100%
H ₂ (4q)	HVA ($d = 2$, 28p)	737.68 $\pm$ 1.5	514.54 $\pm$ 25	†	100%
H ₄ (8q)	LPTE (TOPAZ)	235.15	83.82	83.75	100%
H ₄ (8q)	HEA (depth 3)	798.39	626.54	626.58	100%
H ₄ (8q)	HVA ($d = 2$, 48p)	113.41 $\pm$ 9.3	53.49 $\pm$ 2.2	†	100%
H ₄ (8q)	HVA ($d = 2$, 368p)	2716.68 $\pm$ 5.8	2282.79 $\pm$ 467	†	100%

*Clamped to FCI; see text for discussion. †ZNE+CDR not run on HVA; the HVA noiseless floor itself already sits far above FCI (see text).

on the best seed — within chemical accuracy. Across 10 seeds the noisy baseline averages 71.96 mHa and ZNE+CDR recovers 26.16 mHa on average, with a noiseless mean of 26.72 mHa — the structural well sits at ~ 25 mHa even under FakeFez noise, and ZNE+CDR recover the full noiseless floor because LPTE’s shallow depth (~ 8 effective two-qubit gates) preserves the noise-energy relationship. The same holds at the 8-qubit scale: on H₄, LPTE’s ZNE+CDR (83.75 mHa) recovers the noiseless floor (83.82 mHa) almost exactly.

The HEA ZNE+CDR entry in Table 6 is marked with an asterisk because the second-order polynomial ZNE extrapolation produced a negative ΔE_{FCI} — a known artifact when the noise curve deviates from the assumed exponential model. We clamp all ZNE and CDR values at FCI before computing statistics. The clamping rate itself is a diagnostic: HEA required clamping on 8 of 10 seeds, while TOPAZ/LPTE required clamping on none. That TOPAZ’s extrapolations never violated the physical bound implies its noise-energy relationship is closer to the assumed exponential form — a direct consequence of the structural well pre-tuning the circuit into a region where the noise model is better-behaved.

The HEA boundary-case artifact and the HVA parameter-sensitivity (Sec. 8.4). Table 6 shows the central finding of the VQE benchmark. On the 4-qubit H₂ toy system HEA marginally out-performs LPTE (53.47 vs. 71.96 mHa noisy) because generic ansätze possess sufficient expressivity to brute-force the ground state — but this is a boundary-case artifact that inverts at the 8-qubit scale, where HEA degrades 15 $\times$ to 798.39 mHa while LPTE degrades only 3.3 $\times$ to 235.15 mHa. HVA behaves differently: its performance is hypersensitive to the number of parameters, not to system size *per se*. A lightly-parameterised HVA (48 parameters) scales well on H₄ (113.41 mHa noisy, 53.49 mHa noiseless — better than LPTE on the same molecule), whereas the physics-scaled HVA (368 parameters) collapses into a barren-plateau-like regime (2716.68 mHa), and on the 4-qubit H₂ system HVA fails to converge below a noiseless floor of several hundred mHa. The resistance of LPTE to this parameter-count pathology — it never runs ZNE+CDR above FCI and its QEM recovery is nearly exact at both scales — is the algorithmic contribution, alongside the MMA-vs.-COBYLA comparison treated in Sec. 8.4.

8.4 Why STANDARD ansätze fail, and what that means for TOPAZ

The combined benchmark exposes the mechanism by which generic variational circuits degrade under FakeFez, and shows that LPTE’s recovery is a structural property of the ansatz rather than an incidental advantage.

The HEA boundary-case artifact. HEA marginally outperforms LPTE on the 4-qubit H_2 toy system (53.47 vs. 71.96 mHa noisy) purely because a depth-3 hardware-efficient circuit has enough free parameters to approximate the two-qubit ground state. This advantage is a boundary artifact of small Hilbert space, not a general property: at the 8-qubit H_4 scale the identical HEA setup degrades $15\times$ to 798.39 mHa while LPTE degrades only $3.3\times$ to 235.15 mHa. The inversion follows from the structural-well picture: generic ansätze are not pre-tuned to any noise *basin*, so their energy landscape has no reason to remain stable under the noise’s deformation, whereas TOPAZ’s optimisation explicitly settles the parameters inside a region that resists the dominant error channels.

HVA: a parameter-count pathology, not a design failure. The HVA results show that naive scaling is the limitation, not the variational-hardware-ansatz (VHVA) principle itself. A lightly-parameterised $d = 2$ HVA with 48 parameters outperforms LPTE on H_4 (113.41 mHa noisy, 53.49 mHa noiseless), because in its low-parameter regime it reduces to a familiar physically-inspired ansatz that the gradient-based solver can optimise. Progressively scaling to the physics-consistent 368-parameter form collapses performance by an order of magnitude (2716.68 mHa) as the parameter space becomes too large for reliable optimisation — a barren-plateau-like signature. On H_2 the 28-parameter HVA likewise stalls at a noiseless floor of ~ 512 – 613 mHa. Generic ansätze thus carry no built-in mechanism to navigate this parameter-count cliff, whereas TOPAZ’s LPTE reduces the effective degrees of freedom via self-pruning and therefore remains on the well side of the cliff at both scales. The collapse of the 368-parameter HVA into a barren-plateau-like regime is a direct manifestation of expressibility-induced barren plateaus [29]: high expressibility concentrates cost-function gradients exponentially in qubit count and depth, producing flat landscapes where gradient-based optimisers stall. LPTE’s self-pruning to 8–24 effective dimensions explicitly curtails expressibility to the noise-robust subspace, which is why it resists this collapse.

8.5 Cross-algorithm: BV and Grover

The Grover result is the most revealing. Earlier work reported Grover as incompatible with TOPAZ because amplitude amplification fails the continuity and convexity criteria. Applying TOPAZ only to the diffusion-oracle implementation — treating the oracle as a PTE landscape while leaving the Boolean search logic untouched — recovers a $3.09\times$ resilience advantage. The principle generalises: non-differentiable algorithms can be protected by acting on their structural primitives.

8.6 $H_4 \rightarrow$ TFIM cross-Hamiltonian transfer

A topology optimiser would be redundant if a single generic noise-resilient ansatz — a pre-trained circuit — could be reused across many physics problems. The cross-Hamiltonian test examines exactly this: we take the optimised Layout A (TOPAZ on the 8-qubit H_4 VQE molecule, 84 topological parameters, 263 entangling gates, noisy $\Delta E_{\text{FCI}} = 220.1$ mHa, noiseless 57.5 mHa) and re-enter it as the ansatz for an 8-qubit TFIM with transverse field at the quantum-critical point,

$J = h = 1$ (FCI $E_0 = -9.837951$ Ha). Two settings are compared:

Transfer (frozen structure). The H_4 -optimised topology is held fixed and only the rotation angles are re-optimised by MMA. The structure was chosen for the XX-, YY-, ZZ-linking geometry of the H_4 Hamiltonian, and TFIM does not share those symmetries. Noisy ΔE_{FCI} averages 1211.0 mHa (best 1144.7, worst 1271.1); noiseless 714.5 mHa.

From-scratch (fresh structure). The full TOPAZ pipeline (topology and angles) is re-run directly on the TFIM target. Noisy ΔE_{FCI} averages 932.1 mHa (best 845.3, worst 989.9); noiseless 436.9 mHa.

The transferred topology is worse by 278.9 mHa on average. This penalty cannot be attributed to depth or gate count, because Layout A (263 gates) and the from-scratch Layout B (264 gates) are essentially identical in size; the gap is therefore purely structural. Even a fresh TOPAZ run leaves a residual ~ 0.4 – 1.2 Ha error on this strongly-correlated critical-point TFIM, but the decisive result is the ordering: tailoring the topology to the explicit symmetries of the target Hamiltonian beats reusing one from a different physics domain. TOPAZ optimises for the target’s noise-resilience and entangling channels together; it does not reduce to a generic reusable circuit.

8.7 Validation on FakeFez

All experiments in this subsection use the 8-qubit Trotter ansatz with $M = 21$ nearest-neighbour Pauli terms on the quietest `ibm_fez` path ([127, 137, 147, 146, 145, 144, 143, 136], path mean two-qubit gate error 0.31%). Optimisation runs against FakeFez; the same (ρ, τ) are then submitted to live hardware.

8.8 Noise-model transfer matrix

Parameters optimised under one noise model were cross-evaluated on five others, producing a $6 \times 6 \times 10$ transfer matrix (360 evaluations). The six models are: nominal (triple-channel), low-noise (50% amplitude), high-depolarisation ($2\times$), amplitude-damping, phase-damping, and high-readout (5%/8%). The critical result is the transfer efficiency: parameters optimised under one noise model perform at 99.4% of their on-model fidelity when evaluated under a different model (Fig. 3). The structural well is not an overfit to a specific noise profile but a generic geometric property of the LPTE ansatz under structured noise. This has a practical corollary: parameter sets optimised in simulation transfer to hardware without re-optimisation.

Noiseless fidelity decomposition. LPTE achieves $F_{\text{ideal}} = 0.9829$ (approximation loss 1.71%); under noise it records $F_{\text{noisy}} = 0.9085$ (noise loss 7.41%). Exact preparation loses 14.99% under identical noise (0.8501 from 1.0). LPTE is roughly $2\times$ more resilient to noise than the exact unitary, and the approximation penalty is only 1.71% of the gap to perfect fidelity. The advantage is dual-mechanism: the structural well absorbs noise more effectively, and the polar projection stabilises optimisation.

Multi-seed stability. Across $n = 10$ independent runs under FakeFez, the optimiser converges to DM 0.739 ± 0.066 , inversion test 0.552 ± 0.044 , and 9.0 ± 1.3 active terms out of

[Figure 3. 6×6 noise-model transfer matrix]

Figure 3. 6×6 noise-model transfer matrix. Transfer efficiency: 0.9943.

21. Every seed prunes to 7–12 active terms — never all 21 — confirming self-pruning is systematic.

8.9 Real hardware: ibm_fez

The optimised (ρ, τ) from the FakeFez benchmark (Sec. 8) are submitted to live `ibm_fez` (Heron r2, 156 qubits). State preservation is measured with a two-way Loschmidt-echo protocol: prepare the optimised LPTE state with U , apply the inverse $U^\dagger$, and measure the return probability. The round trip runs twice the forward circuit plus the transpilation overhead of the target inverse, so it is a strictly harder experiment than single-pass preparation and isolates state preservation from measurement-operator structure. The quietest 8-qubit path is [125, 124, 123, 136, 143, 142, 141, 140]; the optimised ansatz uses 19 two-qubit gates at depth 71 — already sparser than the 21-gate, depth-82 reference target it approximates.

The two-way Loschmidt-echo bound. Under depolarizing noise, the round-trip echo obeys a square law. Let a single forward pass through U accumulate an effective two-qubit error rate ε per gate over n entangling gates, preserving $F_{1\text{-way}} \approx 1 - n\varepsilon$. The round trip $U^\dagger U$ runs twice the entangling layers (the inverse is not recompiled back into the forward form), and the two passes accumulate independent noise, giving

$$F_{\text{echo}} \approx 1 - 2n\varepsilon \approx (1 - n\varepsilon)^2 = F_{1\text{-way}}^2. \quad (17)$$

For the measured one-way fidelity $F_{1\text{-way}} = 0.5758$, this bounds the ideal round-trip echo at $F_{\text{echo}} \approx (0.5758)^2 \approx 0.3315 = 33.15\%$. The measured raw echo of $17.27\% \pm 0.60\%$ attains 52% of that ceiling; the residual drop is accounted for by the CNOT-unrolling depth of $U^\dagger$ (which more than doubles the entangling layers relative to the forward pass) and by eight-qubit spectator crosstalk along the path — both effects invisible to the single-pass fidelity. The raw figure uses no error-mitigation post-processing.

The failure of standard ZNE. ZNE fails, and this failure is diagnostic. Gate-stretching for noise amplification pushes the already-deep two-way circuit beyond the device’s coherence window, and the extrapolation collapses to 5.78% — below the raw echo and far below the 33.15% square-law ceiling. Tensor-product noise-model (TPNM) correction records 18.20%, marginally above raw. The pre-execution paradigm is confirmed precisely by this contrast: TOPAZ’s internal ansatz optimisation succeeds where post-processing extrapolation fails, because it acts on the circuit before noise accumulates rather than trying to extrapolate through the noise.

Relative to the standard product-Trotter baseline (0.0155 ± 0.0028), the raw TOPAZ echo (0.1727 ± 0.0060) is the $11.14\times$ hardware gain at 18σ quoted in the abstract ($0.1727/0.0155 = 11.14$; 1014% increase). Because the baseline and the TOPAZ circuit suffer identical real-hardware overhead proportionally, the gain ratio — rather than the absolute fidelity — is the cross-backend metric. Total QPU time was ~ 2 minutes 15 seconds.

Why the echo is lower than one-way DM. The two-way echo runs $2\times$ the entangling depth, so by Eq. (17) it inherits the square of the one-way fidelity rather than a linear fraction. The gap between the 33.15% square-law ceiling and the measured 17.27% decomposes into the CNOT-unrolling overhead of $U^\dagger$ ($\sim$ most of the residual) and spectator crosstalk along the 8-qubit path; the residual below the raw figure is then explained by the finite shot budget. The one-way DM fidelity itself (57.58%) transfers to hardware with negligible loss, matching the noiseless simulator reference (0.5758 vs. 0.5758).

8.10 Stacking with ZNE, CDR, VD

TOPAZ composes multiplicatively with post-processing techniques because they act on different parts of the pipeline: TOPAZ pre-tunes parameter space; ZNE, CDR, and VD extrapolate or correct observables. The relevant comparison is not “TOPAZ vs. ZNE” but “what does the stack look like”. ZNE alone at the baseline gives marginal improvement ($0.0677 \rightarrow 0.0747$). On top of TOPAZ, ZNE contributes a further 0.0667 DM gain. TOPAZ moves the operating point onto a part of the noise curve where ZNE’s extrapolation is more reliable.

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